

Current Research Progress

Personalized Sign Language To Voice Conversion

Current milestone: A localized Sinhala dataset of approximately 1,400 samples has been created. The samples include active SLSL video clips, extracted skeletal features, Sinhala captions, aligned Sinhala audio, and timing information.

1. MediaPipe Holistic feature representation

Skeletal and visual landmarks were extracted from each video frame using MediaPipe Holistic. Instead of using every image pixel directly, the system converts the detected body, face, and hand landmarks into numerical coordinates. Under the current extraction configuration, each frame is represented by a 1,662-dimensional feature vector.

- Pose: 33 landmarks x 4 values (x, y, z, and visibility) = 132 values.
- Face: 468 landmarks x 3 coordinates (x, y, and z) = 1,404 values.
- Left hand: 21 landmarks x 3 coordinates = 63 values.
- Right hand: 21 landmarks x 3 coordinates = 63 values.
- Total per frame: $132 + 1,404 + 63 + 63 = 1,662$ values.

Example: if an active-signing clip contains 100 frames, its skeletal-feature sequence has a shape of approximately $100 \times 1,662$. The model processes the ordered sequence of frame-level vectors to learn changes in body pose, hand shape, hand movement, and facial expression over time.

Future modification note: The 1,662-value vector describes the current full MediaPipe Holistic configuration; it is not a permanent project limitation. During later experiments, the feature representation may be modified by removing or reducing facial landmarks, selecting only important joints, adding derived features such as joint angles, distances or velocities, normalizing coordinates, handling missing landmarks, or combining skeletal features with RGB/video features. Any such modification will change the feature dimension and must be recorded with the corresponding model results.

2. SRT-based Sinhala caption and audio extraction

Sinhala language data was obtained from the caption track associated with each YouTube source video. The video URL was used to retrieve its subtitles as an SRT file. Each SRT entry supplied Sinhala caption text together with a start and end timestamp. The timestamps of the generated active-signing clips were matched against the SRT intervals, and the corresponding portions of the original Sinhala audio were cropped. Each dataset sample therefore links the active video, Sinhala caption, Sinhala audio, skeletal features, and timestamp metadata.

3. Existing research-model training with How2Sign

Before developing the Sinhala model, an existing sign-language research model was trained and evaluated using the How2Sign dataset. Prediction outputs and accuracy values were obtained from the trained model. This experiment confirmed that the data-loading, feature-processing, training, prediction, and evaluation pipeline could operate correctly, and it provided a working reference for developing the Sinhala model. The exact prediction and accuracy values should be inserted into the final experimental-results section when the verified results are available.

4. Mid-sentence cropping issue and tested settings

Some active-signing videos were cropped in the middle of a sentence. Motion pauses do not always represent sentence endings, and a fixed time window can divide a continuous sentence. Different segmentation settings were therefore examined.

- An early fixed-duration configuration generated 5-second clips with 50% overlap.
- The later adaptive configuration used a motion threshold of 0.01, a minimum signing duration of 2 seconds, a minimum idle duration of 1 second, and a smoothing value of 10.
- Maximum clip duration was set to 18 seconds, with 1-second overlap, no start padding, and 1.2 seconds of end padding.
- These adjustments improved clip completeness, but some sentence boundaries still require merging with neighbouring clips or manual correction using SRT timestamps and caption punctuation.

5. Current project status

Dataset: Approximately 1,400 Sinhala samples have been prepared.

Features: The current samples contain 1,662-value MediaPipe Holistic feature vectors for each frame; this feature layout may be modified in future experiments.

Language alignment: Sinhala captions and audio have been extracted through SRT and active-clip timestamp alignment.

Reference training: An existing research model has been trained using the How2Sign dataset, and prediction outputs and accuracy values have been obtained.

Remaining data issue: Some clips begin or end in the middle of a sentence and require final alignment correction.

Sinhala translation model: Training on the localized Sinhala dataset has not yet been completed.

6. Next stage: Sinhala-model development

The next stage is to develop and train a model specifically for the approximately 1,400-sample Sinhala dataset. Before training, the video, current 1,662-value skeletal sequence, Sinhala caption, Sinhala audio, and timestamps must be checked for one-to-one alignment. Mid-sentence clips should be merged, extended, corrected, or removed. The verified dataset should then be divided into training, validation, and test sets, after which the existing research-model pipeline tested with How2Sign can be adapted for Sinhala output. Further experiments may compare the full 1,662-value representation with reduced or enhanced feature configurations. Each model should be evaluated using prediction quality, translation accuracy, BLEU score, and processing latency.

Current progress statement

At present, the project has nearly completed the main dataset-construction and reference-model validation stages, although a few final touch-ups remain. Approximately 1,400 Sinhala samples have been created with MediaPipe Holistic features, SRT-based Sinhala captions, and timestamp-aligned audio. An existing research model has been trained using the How2Sign dataset, and prediction outputs and accuracy values have been obtained. The immediate tasks are to complete and finalize the dataset and then train a translation model using the Sinhala dataset.